# TOPFAITH UNIVERSITY
## MKPATAK, AKWA IBOM STATE, NIGERIA

### B.Sc. DEGREE EXAMINATION

**FACULTY:** COMPUTING AND APPLIED SCIENCES **SESSION**: 2023/2024
**DEPARTMENT:** DEPATMENT OF COMPUTING **SEMESTER**: PRIMO
**COURSE CODE**: TUM-CSC112 **CREDIT UNIT**: 2
**COURSE TITLE**: WEB TECHNOLOGIES 1 **TIME**: 2 HOURS
**INSTRUCTION**: ANSWER QUESTION 1 AND ANY OTHER TWO (2) QUESTIONS

1. (a) Write an HTML code for the following components in the table below: (8 marks)

| S/N | Expression | Answer |
|---|---|---|
| 1 | An ok button | |
| 2 | A name field | |
| 3 | A password field | |
| 4 | A dropdown list | |

(b) Define by example, the complete structure of an HTML document (8 marks)

(c) In a tabular form, state the output of the following expression, given that a = 4 and b = 4. (7 ½ marks)

| S/N | Expression | Answer |
|---|---|---|
| 1 | a != b | |
| 2 | a % b | |
| 3 | (a < b)? a:b | |
| 4 | a < a && a == a | |

**[Total = 23½ marks]**

2. (a) Write a CSS class to style the components in the table described below: (8 marks)

| S/N | Expression | Answer |
|---|---|---|
| 1 | A green colored label text | |
| 2 | A circular profile picture avatar | |
| 3 | A box with red-colored shadow | |
| 4 | A button that grows on mouse over | |

(b) Using a flexbox approach, write a CSS style to perfectly centralize an image in oval-shaped div component (8 marks)

(c)Programmatically show the relationship between the following files – (1) style.css (2) actions.js (3) page.html (7½ marks)

**[Total = 23½ marks]**

3. (a) Define suitable CSS selectors for the HTML custom components in the table below: (8 marks)

| S/N | Expression | Answer |
|---|---|---|
| 1 |  …  | |
| 2 |  …  | |
| 3 |  …  | |
| 4 |  …  | |

(b) State the output of the following expression given that a = 10, b = 20. (8 marks)

| S/N | Expression | Answer |
|---|---|---|

| | | |
|---|---|---|
| 1 | a += b | |
| 2 | a %= b | |
| 3 | a *= 2 | |
| 4 | (a--)+1 | |

(c)Using CSS, define a div component of size 200px by 200px. Transform the div so that the width will be twice as large without modifying the initial width property. (7½ marks)

**[Total = 23½ marks]**

4. (a) Write a Javascript function to change the value of a div component with id=”div” to “Topfaith University” (11 ½ marks)

(b) Write a Javascript function to find white eggs in a basket of multi-colored eggs. Eggs are stored as strings (12marks)

**[Total = 23½ marks]**

5. (a) Given the loop construct below: (12 marks)

```
for(var i=0; i<=N; i++){
    …
}
```

In a tabular form, answer the questions below:

| S/N | Expression | Answer |
|---|---|---|
| 1 | How many times will the loop’s body get executed | |
| 2 | At what value of i will the loop terminate | |
| 3 | What will happen if i++ is replaced with i-- | |

(c) Using HTML, design an OTP confirmation page comprising of a page header, OTP input field, and a confirm button all centralized. (11½ marks)

**[Total = 23½ marks]**

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**COURSE CODE**: TUM-CSC112
**COURSE TITLE**: WEB TECHNOLOGIES 1
**COURSE COORDINATOR**: EMMANUEL E. NYOHO
**COURSE LECTURERS:** EMMANUEL E. NYOHO

**SESSION**: 2023/2024
**SEMESTER**: PRIMO
**CREDIT UNIT**: 2
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## MARKING GUIDE

QUESTION No. 1

1. (a) Write an HTML code for the following components (8 marks)
**[2 marks each]**

| S/N | Expression | Answer |
|---|---|---|
| 1 | An ok button | <button>Ok</button> |
| 2 | A name field | <input type="text" /> |
| 3 | A password field | <input type="password" /> |
| 4 | A dropdown list | <select><br><option>…</option><br>…<br></select> |

(b) Define by example, the complete structure of an HTML document (8 marks)
**[2 marks each]**

**Answer:**
<!DOCTYPE HTML>
<html>
<head>
<title>…</title>
</head>
<body>…</body>
</html>

(c) State the output of the following, given that a = 4 and b = 4 (7 ½ marks)
**[2 marks each, 1 ½ mark for (1)]**

| S/N | Expression | Answer |
|---|---|---|
| 1 | a != b | False |
| 2 | a % b | 0 |
| 3 | (a < b)? a:b | b |
| 4 | a < a && a == a | False |

**[Total = 23½ marks]**

QUESTION No. 2

2. (a) Write a CSS class to style the components described below: (8 marks)
**[2 marks each]**

| S/N | Expression | Answer |
|---|---|---|

| | | |
|---|---|---|
| 1 | A green colored label text | .label{color:green;} |
| 2 | A circular profile picture avatar | .avatar{<br>Border-radius:50%;<br>} |
| 3 | A box with red-colored shadow | .box{<br>Box-shadow: 0 0 10px 20px red;<br>} |
| 4 | A button that grows on mouse over | .button:hover{<br>Transform:scale(1.1,1.1);<br>} |

(b) Using a flexbox approach, write a CSS style to perfectly centralize an image in oval-shaped div component (8 marks)

**[2 marks each style rule]**

**Answer:**

```
.center{
Display:flex;
Justify-content:center;
Align-items:center;
}

.div{
Border-radius:50%;
}
```

(c)Programmatically show the relationship between the following files – (1) style.css (2) actions.js (3) page.html (7½ marks)

**[3 marks each link and script tag, 1 ½ mark for html, head construct]**

**Answer:**

```
<html>
<head>
<link rel=”stylesheet” href=”style.css”/>
<script src=”actions.js”></script>
</head>
…
```

**[Total = 23½ marks]**

QUESTION No. 3

3. (a) Define suitable CSS selectors for the following HTML custom components. (8 marks)
**[2 marks each]**

| S/N | Expression | Answer |
|---|---|---|
| 1 |  …  | .class{} |
| 2 |  …  | #identity{} |
| 3 |  …  | div{} |
| 4 |  …  | span{} |

(b) State the output of the following expression given that a = 10, b = 20. (8 marks)
**[2 marks each]**

| S/N | Expression | Answer |
|---|---|---|
| 1 | a += b | 30 |
| 2 | a %= b | 10 |
| 3 | a *= 2 | 20 |
| 4 | (a--)+1 | 10 |

(c)Using CSS, define a div component of size 200px by 200px. Transform the div so that the width will be twice as large without modifying the initial width property. (7½ marks)

**[2 marks for each style rule, 1 ½ mark for div{}]**

**Answer:**

```
div{
width:200px;
height:200px;
transform:scaleX(2);
}
```

**[Total = 23½ marks]**

QUESTION No. 4

4. (a) Write a Javascript function to change the value of a div component with id=”div” to “Topfaith University” (11 ½ marks)

**[5 marks each for function implementation, 1 ½ mark for function definition]**

**Answer:**

```
Function changeText(){
    Var component = document.getElementById(“div”);
    Component.innerHTML = “Topfaith University”;
}
```

(b) Write a Javascript function to find white eggs in a basket of multi-colored eggs. Eggs are stored as strings (12marks)

**[6 marks for each for loop and if statement]**

**Answer:**

```
Var foundAt = -1;
For(var i=0; i<eggs.length; i++){
    If(eggs[i] ==”white”){
            foundAt = I;
            return;
    }

}
```

**[Total = 23½ marks]**

QUESTION No. 5

5. (a) Given the loop construct below: (12 marks)

**[4 marks each]**

```
for(var i=0; i<=N; i++){
    …
}
```

| S/N | Expression | Answer |
|---|---|---|
| 1 | How many times will the loop’s body get executed | 11 |
| 2 | At what value of i will the loop terminate | 11 |
| 3 | What will happen if i++ is replaced with i-- | infinite |

(c) Using HTML, design an OTP confirmation page comprising of a page header, OTP input field, and a confirm button all centralized. (11½ marks)

**[2 marks for each tag head, center, h3, input, button, 1 ½ mark for html tag]**

**[Total = 23½ marks]**

**Answer:**

```
<html>
<head> <title> … </title> </head>
<body>

<center>
<h3> heading </h3>
<input type=”text” />
<button>Confirm</button>
</center>
</body>
</html
```